Exercises Week 13

Tuesday, November 12, 2024 12:39 PM

PG 4

Undirected:

{1, 5, 4} = 2

{1, 2, 4} = 2

{1, 2, 3, 4} = 3

Directed:

{1, 2, 4} = 2

PG 8

1. It's an array of linked lists. Each node on the graph is represented as an index in the array, and each edge that is in the graph is represented as a node in the linked list from the starting vertex to the ending one, where the starting vertex is the index of the array, and the ending one is a node in the linked list.

2. Array and linked list…?

3. The missing edge is between 3 and 5.

Pg 9:

Out:

A 2 E 0

B 0 F 3

C 2 G 1

D 3

Pg 10: $V = \{1, 2, 3, 4, 5, 6, 7\}$

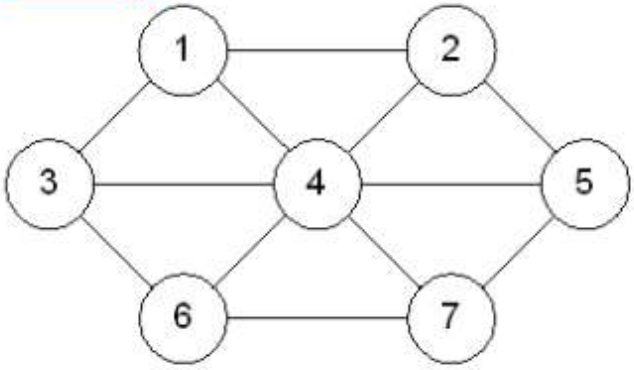
$E = \{(1,2), (1,3), (1,4), (2,4), (2,5), (3,4), (3,6), (4,5), (4,6), (4,7), (5,7), (6,7)\}$

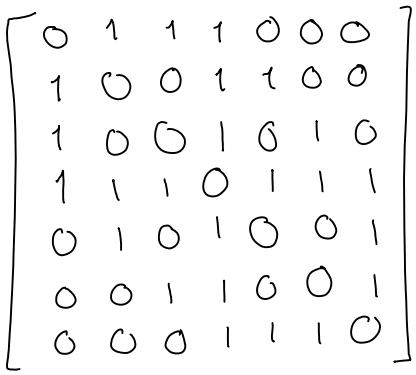
Vindegree: $\{1:3, 2:3, 3:3, 4:6, 5:3, 6:3, 7:3\}$

Voutdegree: $\{1:3, 2:3, 3:3, 4:6, 5:3, 6:3, 7:3\}$

1. Show how the **undirected** graph is represented using an ***adjacency matrix*** (2D array).

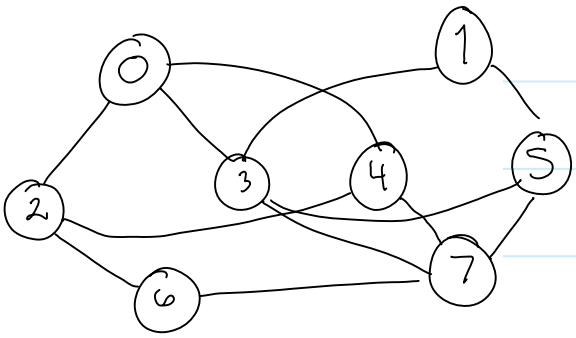
Answer$_b$:





2. Convert the adjacency matrix below to an **undirected** graph. **Answer$_b$:**

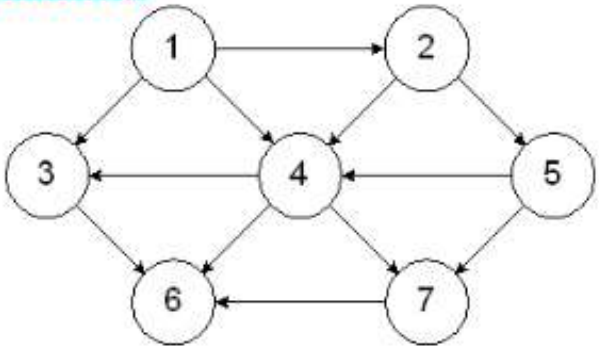
	0	1	2	3	4	5	6	7
0			1	1	1			
1				1		1		
2	1				1		1	
3	1	1				1		1
4	1		1					1
5		1		1				1
6			1					1
7				1	1	1	1	



*consistent graph drawling?

3. Consider the **directed** graph shown below.
   a. Show how the graph is represented using matrix.

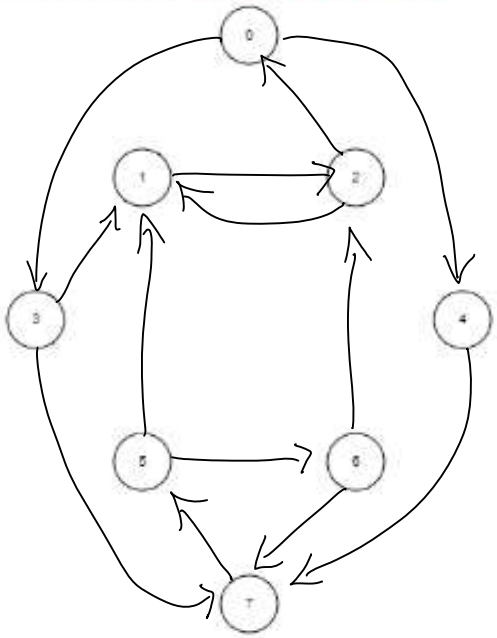
Answer$_b$:



$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

4. Convert the adjacency matrix below to a **directed** graph. **Answer$_b$:**

	0	1	2	3	4	5	6	7
0				1	1			
1			1					
2	1	1						
3		1						1
4								1
5		1					1	
6			1					1
7						1		



** we will use this graph on the test!

5. Based on a matrix, how can you tell if the graph is directed or undirected?

Answer$_b$:

Directed bc its not symetric.